

Max Schulz

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Education

- 2020 - 2024** **M. Sc. Biology**
University Leipzig, Germany
- Thesis in acoustic and computational neuroscience, "AI-Supported Evidence For Spectro-Temporal Primacy in Auditory Numerosity Judgment"
 - Tutoring how to prepare, conduct and analyze psychoacoustic electroencephalogram (EEG) experiments with Python
 - Implementation of custom-written platform which enables asynchronous and simultaneous operation of multiple devices (cameras and processors mainly) via a Python interface (AG Neurobiology)
 - Development of a custom analysis tool which enables EEG-analysis in MNE-Python (AG Neurobiology)
- 2017 - 2021** **B. Sc. Biology**
University Leipzig
- Bachelor Thesis in auditory neuroscience, "Measuring Auditory Tuning Curves With EEG"
- 2016 - 2017** **B. Sc. Biology**
Justus-Liebig University Gießen

Experience

- 2024 - now** **PhD Candidate**
University of Lübeck, Germany
- Researching subprocesses of auditory selective attention, specifically delineating target enhancement from distractor capture and suppression.
 - Utilizing EEG and webcam-based eye and head tracking to study neuro-cognitive mechanisms in young and older adults and infants.
 - Supervised by Dr. Malte Wöstmann and Prof. Dr. Jonas Obleser in the Auditory Cognition Lab.
- 2024** **Student Assistant**
Max-Planck Institute for Human Cognitive and Brain Sciences
- Assisting in the implementation of electrical cortical field simulations for optimization of transcranial magnetic stimulation approaches with PyNibs (Prof. Dr. Gesa Hartwigsen).
- 2021 - 2024** **Student Assistant**
Max-Planck Institute for Human Cognitive and Brain Sciences
- Assisting in study preparation, conduction, and analysis in the Psychology Department (Prof. Dr. Christian Doeller).
 - Conducting behavioral, eye-tracking, virtual-reality, and fMRI studies.

Open Source Projects

- 2020 - now** **mxschlz**
My GitHub repository, mostly containing Python containers for experiment and psychological analysis scripts.

Academic Interests

- Selective Attention
- Auditory Scene Analysis
- Electroencephalography (EEG)

Talks & Symposia

- June 2026** **Symposium Chair & Speaker**
Psychologie und Gehirn (PuG) 2026, Heidelberg, Germany
• {'Organized and chaired the symposium "Dynamics of Enhancement and Suppression': 'Integrating Neural and Behavioral Perspectives on Selective Attention'"}

March 2026 **Symposium Speaker**
Tagung experimentell arbeitender Psychologen (TeaP) 2026, Tübingen, Germany
• Held a talk titled "Behavioral and neural signatures of extracting regularities from relevant and irrelevant sequences of sound"

Jan. 2026 **Invited Talk**
Marlene Meyer's Lab, Donders Institute, Nijmegen, Netherlands
• Presented research on the infant OCAPI (Objective Coding of Attention Pipeline for Infants) project

Skills

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| Programming | <ul style="list-style-type: none">• Python: PsychoPy, MNE-Python, PyNibs, numpy, scipy• Bash / Git• R |
| Research & Methods | <ul style="list-style-type: none">• Electroencephalography (EEG)• Webcam-based Eye & Head Tracking• Psychoacoustics• Virtual Reality (VR) |